

Exercise 11

List all predicates in the twoPred() method. Index them by the line numbers in the program listing.
Write the complete truth table and the definitional method for each predicate.
(A) List all pairs of rows from your table that satisfy predicate coverage (PC) with respect to each clause.
(B) Find values for each predicate to satisfy predicate coverage (PC).
(C) List all pairs of rows from your table that satisfy clause coverage (CC) with respect to each clause.
(D) Find values for each predicate to satisfy clause coverage (CC).
(E) List all pairs of rows from your table that satisfy Correlated Active Clause Coverage (CACC) with respect to each clause.
(F) Find values for each predicate to satisfy correlated active clause coverage (CACC).

```
public static String twoPred (int x, int y){  
  
1. boolean z;  
2. if (x < y)  
3. z = true;  
4. else  
5. z = false;  
6. if (z && x+y == 10)  
7. return "A";  
  
8.else  
  
1. return "B";  
  
}
```

Exercise 11 ΓÇô Logic Coverage on twoPred()

Predicates

There are two predicates in twoPred() :

#	Line	Predicate	Clauses
p1	2	x < y	a = (x < y)
p2	6	z && x+y == 10	a = z , b = (x+y == 10)

Truth Tables

Predicate p1 = a (line 2)

Row	a (x < y)	p1
1	T	T
2	F	F

Predicate $p2 = a \wedge b$ (line 6)

Row	a (z)	b (x+y=10)	p2
1	T	T	T
2	T	F	F
3	F	T	F
4	F	F	F

Definitional Method (when each clause determines the predicate)

$p1 = a$

- $p_a = p1(a=T) \wedge p1(a=F) = T \wedge F = \text{true}$ (a always determines p1).

$p2 = a \wedge b$

- $p_a = p2(a=T) \wedge p2(a=F) = (T \wedge b) \wedge (F \wedge b) = b \wedge F = b$
 $\nexists$ clause a determines p2 when $b = T$.
- $p_b = p2(b=T) \wedge p2(b=F) = (a \wedge T) \wedge (a \wedge F) = a \wedge F = a$
 $\nexists$ clause b determines p2 when $a = T$.

(A) Row pairs satisfying Predicate Coverage (PC)

PC requires each predicate to evaluate both T and F.

- p1:** (Row 1, Row 2)
- p2:** (Row 1, Row 2), (Row 1, Row 3), (Row 1, Row 4)

(B) Values that satisfy PC

Choose tests so each predicate hits T and F.

Test	x	y	p1 (x<y)	z	p2 (z $\wedge$ x+y=10)	return
T1	1	9	T	T	T (1+9=10)	"A"
T2	5	3	F	F	F	"B"

p1: T (T1), F (T2). p2: T (T1), F (T2). $\nexists$ PC satisfied for both.

(C) Row pairs satisfying Clause Coverage (CC)

CC requires every clause to take both T and F.

p1 (single clause a):

- Pair: (Row 1, Row 2)

p2 $\nexists$ clause $a(z)$ (a=T in rows 1,2; a=F in rows 3,4):

- (1,3), (1,4), (2,3), (2,4)

p2 $\nexists$ clause $b(x+y=10)$ (b=T in rows 1,3; b=F in rows 2,4):

- (1,2), (1,4), (3,2), (3,4)

(D) Values that satisfy CC

Each clause needs both T and F.

Test	x	y	$x < y$ (a of p1)	z (a of p2)	$x + y = 10$ (b of p2)
T1	1	9	T	T	T
T2	5	3	F	F	F

- p1.a: T, F $\Gamma \Delta$
- p2.a: T, F $\Gamma \Delta$
- p2.b: T, F $\Gamma \Delta$

(E) Row pairs satisfying CACC

For each major clause, the minor clauses must make it **active** (determine the predicate) and the major clause flips between the two rows. Under CACC, the predicate is allowed/expected to change too (here it must, since $a \Gamma \Delta b$ has only one cofactor).

p1 (only one clause; CACC = PC = CC):

- (Row 1, Row 2)

p2 $\Gamma \Delta$ clause a is active when $b = T$ (rows 1 and 3):

- (Row 1, Row 3) [a flips $\Gamma \Delta \Delta$, p2 flips $\Gamma \Delta \Delta$]

p2 $\Gamma \Delta$ clause b is active when $a = T$ (rows 1 and 2):

- (Row 1, Row 2) [b flips $\Gamma \Delta \Delta$, p2 flips $\Gamma \Delta \Delta$]

(F) Values that satisfy CACC

We need:

- p1: a row with $x < y$ true and one with $x < y$ false.
- p2 clause a active ($b = T$, i.e. $x + y = 10$): one with $z = T$, one with $z = F$.
- p2 clause b active ($a = T$, i.e. $z = T$, i.e. $x < y$): one with $x + y = 10$, one with $x + y \neq 10$.

Test	x	y	p1 row	p2 row	Notes (active clause)	return
T1	1	9	1 (T)	1 (TT)	covers a-active & b-active T	"A"
T2	5	5	2 (F)	3 (FT)	a-active, a flips $\Gamma \Delta \Delta$ CACC for a	"B"
T3	1	2	1 (T)	2 (TF)	b-active, b flips $\Gamma \Delta \Delta$ CACC for b	"B"

Verification:

- **p1 CACC:** T1 gives $a = T$, T2 gives $a = F$. $\Gamma \Delta$
- **p2 CACC for a:** T1 ($a = T, b = T, p2 = T$) and T2 ($a = F, b = T, p2 = F$); $b = T$ in both, a flips, p2 flips. $\Gamma \Delta$
- **p2 CACC for b:** T1 ($a = T, b = T, p2 = T$) and T3 ($a = T, b = F, p2 = F$); $a = T$ in both, b flips, p2 flips. $\Gamma \Delta$